

Micro 4: KKT Validity, Revealed Preference, Aggregation, and Welfare Measures

Session 1: KKT Points, Concave Programming, Equality Constraints, and Constraint Qualification

1. KKT points and solutions need not coincide

(1) KKT points may not be solutions.

Example (with $p_1 = p_2 = m = 1$):

$$\max_{x_1, x_2} x_1^2 + x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

The KKT analysis yields three KKT points:

$$(0.5, 0.5), (1, 0), (0, 1),$$

with respective objective values:

$$0.75, 1, 1.$$

Hence $(0.5, 0.5)$ is a KKT point but not a maximizer.

(2) Solutions may not be KKT points.

Example:

$$\max_{x_1, x_2} -x_2 \quad \text{s.t.} \quad (x_1 - 1)^2 + x_2^2 \leq 1, \quad (x_1 + 1)^2 + x_2^2 \leq 1.$$

Only $(0, 0)$ satisfies both constraints, so $(0, 0)$ is a solution. However, it cannot be a KKT point here (the standard KKT system fails because the required constraint qualification does not hold at the solution).

2. Concave programming: when KKT is sufficient

Consider

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_j(x) \geq 0, \quad j = 1, \dots, m.$$

If the objective f and all constraints $g_1, \dots, g_m$ are **concave**, then the problem is a **concave programming problem** and **any KKT point is a (global) solution**. This is the key condition under which “KKT $\Rightarrow$ optimal” is guaranteed.

3. Equality constraints and the Lagrangian multiplier method

For an equality constraint $g(x) = 0$, rewrite it in standard inequality form:

$$g(x) \geq 0, \quad -g(x) \geq 0.$$

The two associated multipliers can be combined into a single **unrestricted** multiplier $\mu \in \mathbb{R}$, yielding the standard Lagrangian multiplier method:

$$\mathcal{L}(x, \mu) = f(x) + \mu g(x),$$

with first-order conditions in (x, μ) and the constraint $g(x) = 0$.

4. Concave programming with equality constraints

If the problem includes an equality constraint $g(x) = 0$, then (in inequality form) both g and $-g$ must be concave for concave programming to apply. This requires g to be **linear**.

5. Constraint qualification (CQ) and why it matters

With differentiable objective/constraints, a solution is a KKT point if it satisfies a suitable **constraint qualification**. A standard CQ is **linear independence of active constraint gradients** (LICQ): the gradients of binding (active) inequality constraints and all equality constraints are linearly independent at the solution. If CQ fails, a solution may exist but need not be characterized by KKT.

6. Practical summary

- An optimization problem may fail to have a solution.
- KKT applies only when the objective and constraint functions are differentiable (as required by the method).
- A KKT point may fail to be a solution; a solution may fail to be a KKT point (without CQ).
- If the problem is concave programming, then KKT points are solutions.
- If a solution satisfies CQ, then it is a KKT point.

Session 2: Revealed Preference, Aggregation, and Welfare Measures

1. Rationality and the revealed preference problem

A consumer is **rational** if their observed choices solve a utility maximization problem (UMP). In practice, utilities are unobserved; we observe price–wealth pairs and chosen bundles.

Example of observed purchases (three goods):

Purchase	p_1	p_2	p_3	m	x_1	x_2	x_3
A	10	10	10	300	10	10	10
B	10	1	2	130	9	25	7.5
C	1	1	10	110	15	5	9

If wealth is not directly observed, it can be inferred via **budget exhaustion** (Walras' law):

$$m = p \cdot x.$$

2. Weak axiom of revealed preference (WARP)

Let observed Marshallian demand be $x(p, m)$. WARP requires: for any two situations (p, m) and $(\tilde{p}, \tilde{m})$,

$$\text{if } p \cdot x(\tilde{p}, \tilde{m}) \leq m, \text{ then } \tilde{p} \cdot x(p, m) \geq \tilde{m}.$$

Intuition: if bundle $x(p, m)$ is chosen when $x(\tilde{p}, \tilde{m})$ was affordable, then at $(\tilde{p}, \tilde{m})$ the chosen bundle cannot strictly dominate the other by also making it affordable in reverse.

3. Strong axiom of revealed preference (SARP)

SARP strengthens WARP by ruling out **cycles** of revealed preference over a list of observations. In the numerical example above, a particular ordering of observations implies a violation of SARP (a revealed-preference cycle), even though WARP is not violated.

4. Aggregate demand and distribution of wealth

With I consumers, individual wealths $(m_1, \dots, m_I)$, and individual Marshallian demands $x_i(p, m_i)$, aggregate demand is

$$X(p; m_1, \dots, m_I) = \sum_{i=1}^I x_i(p, m_i).$$

In general, X depends on the **distribution** of wealth, not only on total wealth $\sum_i m_i$.

5. Gorman form (when only aggregate wealth matters)

A necessary and sufficient condition for aggregate demand to depend only on **aggregate wealth** is that each consumer's indirect utility is of the **Gorman form**:

$$v_i(p, m_i) = a_i(p) + b(p) m_i,$$

where the slope term $b(p)$ is common across consumers. Using Roy's identity, this condition implies equal marginal propensities with respect to wealth across consumers, which ensures aggregation without dependence on distribution.

6. Welfare evaluation of price changes: EV and CV

Suppose prices change from p to $\tilde{p}$ while wealth is m . Let

$$u_0 = v(p, m), \quad u_1 = v(\tilde{p}, m),$$

and let $e(p, u)$ denote the expenditure function.

(1) **Equivalent variation (EV).** Defined by

$$v(p, m + \text{EV}) = v(\tilde{p}, m) = u_1.$$

By duality,

$$m + \text{EV} = e(p, u_1) \quad \Rightarrow \quad \text{EV} = e(p, u_1) - e(p, u_0).$$

Interpretation: money (at old prices) equivalent to the price change.

(2) **Compensating variation (CV).** Defined by

$$v(\tilde{p}, m - \text{CV}) = v(p, m) = u_0.$$

By duality,

$$m - \text{CV} = e(\tilde{p}, u_0) \quad \Rightarrow \quad \text{CV} = e(\tilde{p}, u_1) - e(\tilde{p}, u_0).$$

Interpretation: money (after the price change) needed to compensate the consumer back to the original utility.

7. EV and CV as integrals (single price change)

If only the price of good 1 changes from p_1 to $\tilde{p}_1$ (other prices fixed at p_{-1}), then by Shephard's lemma:

$$\text{EV} = \int_{p_1}^{\tilde{p}_1} h_1(t, p_{-1}, u_1) dt, \quad \text{CV} = \int_{p_1}^{\tilde{p}_1} h_1(t, p_{-1}, u_0) dt,$$

where $h_1(\cdot)$ is the Hicksian (compensated) demand for good 1.